Multilinear Interpolation

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Contents

1 Questions for Thiele

- how/what kind of multilinear interpolation is used in the multiple ergodic averages
- how important are the bad tuples (quasi banach estimates) in practice? do we need them formalized?
- how important is it to have the optimal bounds on the weak parameters of the multilinear interpolation? does a weaker bound suffices if it means an easier proof?

2 Introduction

These notes offer an overview of two approaches to multilinear interpolation. They are framed in the setting of [1, Chapter 2], which provides explanation for the notation and definitions used here and in [2].

The argument is split in two cases.

2.1 Interpolating between two points

This is the case considered in [2]. It also seems to be used as a last step in the convex hull case, to be covered on section 2.2. Let

$$\overline{A}_1 = (A_0^{(1)}, A_1^{(1)}), \ldots, \overline{A}_n = (A_0^{(n)}, A_1^{(n)})$$

and $\overline{B} = (B_0, B_1)$ be Banach couples, and let

$$T\colon \Sigma(\overline{A}_1) \times \cdots \times \Sigma(\overline{A}_n) \to \Sigma(\overline{B}),$$

be a multilinear function that restricts to two multilinear functions

$$T\colon A_i^{(1)} \times \cdots \times A_i^{(n)} \to B_i$$

for $i = 0, 1$, such that there are constants K_0, K_1 for which

$$\|T(f_1, \ldots, f_n)\| \leq K_i \|f_1\| \cdots \|f_n\|$$

for $f_k \in A_i^{(k)}$ for $1 \leq k \leq n$, and each $i = 0, 1$.

Given $\theta \in (0, 1)$ and numbers $q_1, \ldots, q_n, q \in [1, \infty]$ satisfying some specified conditions, we would be interested in showing that T also makes sense as a bounded multilinear operator between the intermediate subspaces given by the real interpolation method:

$$T\colon (\overline{A}_1)_{\theta,q_1} \times \cdots \times (\overline{A}_n)_{\theta,q_n} \to (\overline{B})_{\theta,q},$$

with some constant K depending on K_1 and K_2.

Example 2.1. If our Banach couple $\overline{A} = (A^{(0)}, A^{(1)})$ is given by Lebesgue spaces $A^{(0)} = L^{p_0,q_0}$, $A^{(1)} = L^{p_1,q_0}$ and $p_0 < p_1$, then the real interpolation space corresponds to the Lebesgue space $L^{p,q} = (\overline{A})_{\theta,q}$, where $p^{-1} = (1-\theta)p_0^{-1} + \theta p_1^{-1}$.

2.2 Interpolating on the interior of a convex set

We assume that the Marcinkiewicz real interpolation theorem for linear operators is available, and we see how it can generalize to the multilinear case.

[3]

3 Applications

3.1 Usage in the commuting ergodic averages paper

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References

[1] Jöran Bergh and Jörgen Löfström. *Interpolation Spaces: An Introduction*. Repr. Die Grundlehren Der Mathematischen Wissenschaften in Einzeldarstellungen 223. Beijing: World Publ, 2003. 207 pp. ISBN: 978-3-540-07875-3 978-7-5062-6011-4 978-0-387-07875-5 (cit. on p. 1).

[2] Svante Janson. "On Interpolation of Multi-Linear Operators". In: *Function Spaces and Applications*. Ed. by Michael Cwikel et al. Red. by A. Dold and B. Eckmann. Vol. 1302. Berlin, Heidelberg: Springer Berlin Heidelberg, 1988, pp. 290–302. ISBN: 978-3-540-18905-3 978-3-540-38841-8. DOI: `10.1007/BFb0078880`. URL: `http://link.springer.com/10.1007/BFb0078880` (visited on 06/11/2026) (cit. on p. 2).

[3] J. -L. Lions and J. Peetre. "Sur Une Classe D'Espaces D'Interpolation". In: *Publications Mathématiques de l'Institut des Hautes Études Scientifiques* 19.1 (Dec. 1, 1964), pp. 5–68. ISSN: 1618-1913. DOI: `10.1007/BF02684796`. URL: `https://doi.org/10.1007/BF02684796` (visited on 06/14/2026) (cit. on p. 2).